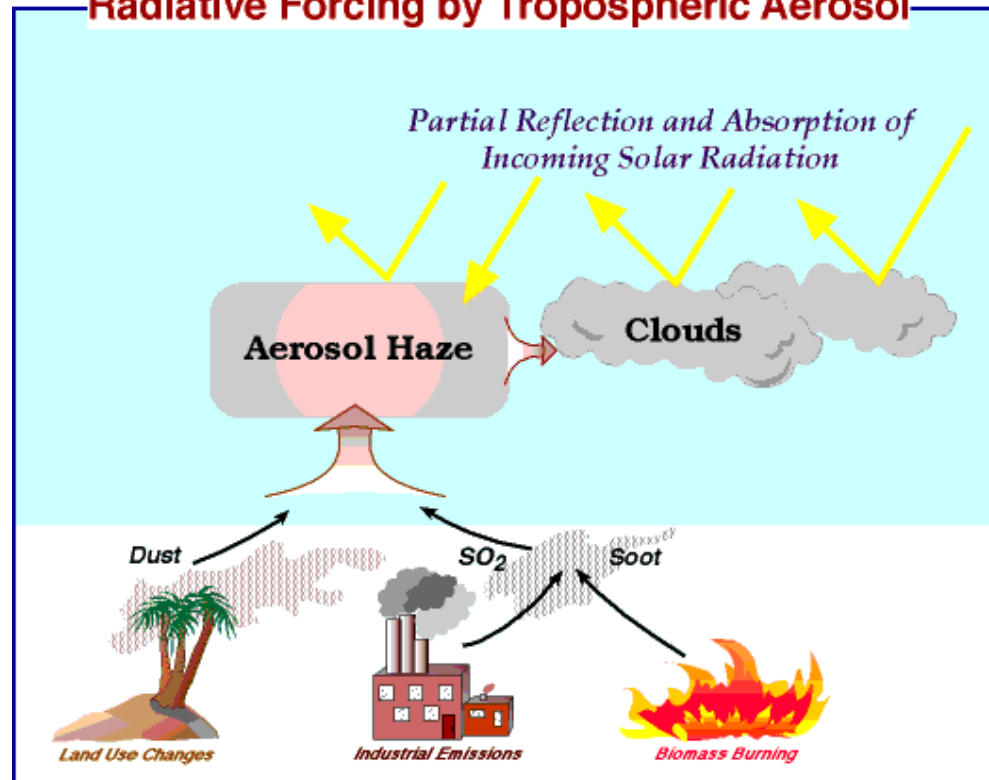


AEROSOLS

Radiative Forcing by Tropospheric Aerosol



What are Aerosols? simply it is one of the pollutant.

✳️ Aerosols are “solid or liquid particles dispersed (suspended) in a gaseous medium.” In the case of atmospheric aerosols, the dispersion medium is the atmosphere.

✳️ Atmospheric particulate matter – also known as particulate matter (PM) or particulates – are microscopic solid or liquid matter suspended in the Earth's atmosphere. The term aerosol commonly refers to the particulate/air mixture, as opposed to the particulate matter alone



Types of Aerosols:

Natural;

Soil Dust, Sea Salt, volcanic Dust, Organic particles, Oceanic sulphates.

Man Made;

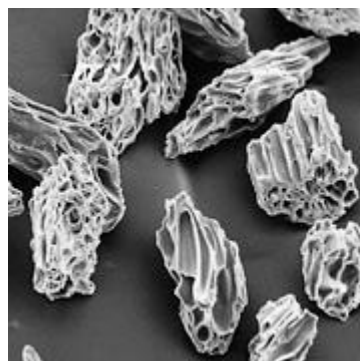
Industrial Sulphates, soot (Black Carbon), Organic Particles.



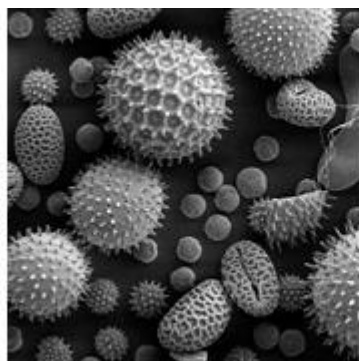
sea salt

dust

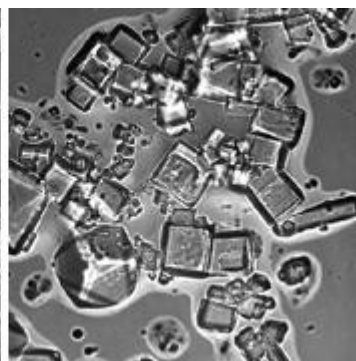
volcanic ash



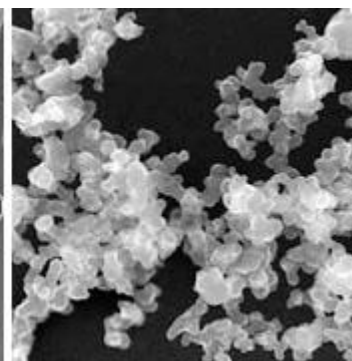
Volcanic
ash



Pollen



Sea salt



Soot

- An aerosol is a collection of suspended particles and the surrounding gases.
- Aerosols are the particles in the solid or liquid phase in gaseous matrix with low settling velocity. They have more important role in atmospheric chemical and physical change, scattering of light.
- A particulate matter refers just to the suspended solid or liquid matter.

Natural sources of aerosols

Desert aerosols



Mineral dust



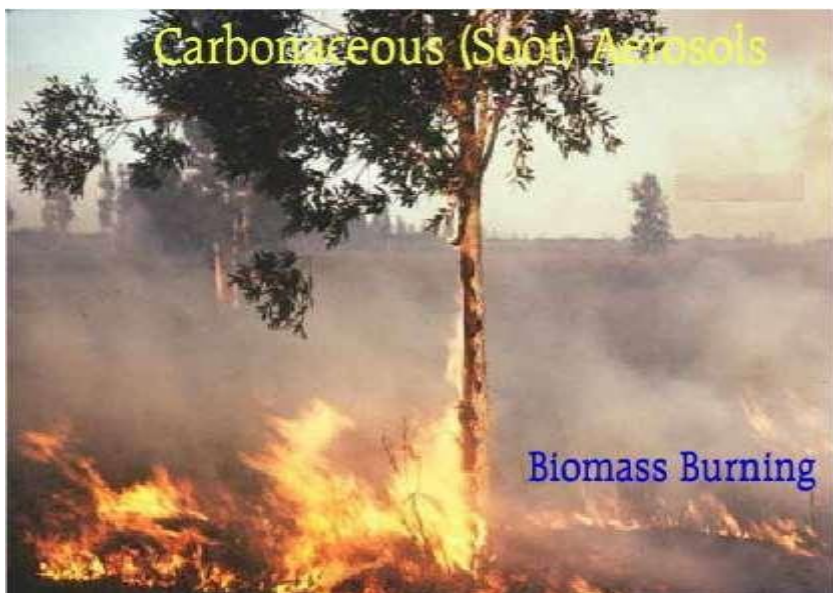
Volcanic aerosols



Marine aerosols



Anthropogenic sources of aerosols



Impact of Aerosols

- ❑ Aerosols affects the climate directly by scattering and absorbing the incoming solar radiation and outgoing infrared radiation in the atmosphere.
- ❑ Aerosols alter the formation and precipitation efficiency of liquid water, ice and clouds, thereby causing an indirect radiative forcing associated with changes in cloud properties.
- ❑ Aerosols act as small atmospheric reactors for heterogeneous chemistry. For example, the strong ozone hole (depletion) observed over antarctica would not take place without aerosols to provide surfaces for heterogeneous reactions.

- ❑ The impact of gaseous and particulate pollutants is gauged by the effect on pollution receptors. For example, deposition of acidic pollutants can speed up the deterioration of building materials, harm vegetation, damage the aquatic ecosystem, cause breathing problems and even increase the mortality rate among humans.
- ❑ Aerosols are the primary cause of visibility degradation in polluted areas, and sometimes even leads to transportation accidents.

Sources of aerosols

Natural (70%)

- Wind blown dust
- Sea-spray
- Volcanic Effluvia
- Forest Fires
- Pollens and vegetation
- Meteoric debris
- Reaction products of gaseous emissions from forests/ oceans

Anthropogenic(30%)

- Industrial and Mining
- Transport
- Biomass and Fossil fuel burning; shifting cultivation
- Urbanization
- Reaction products from gaseous effluvia of human activities

Anthropogenic Sources

Industrial & automobile emissions, Fossil Fuel and Biomass Burning, Agricultural practices, Urbanisation, Anthropogenically enhanced natural sources (eg: forest fires, land-use changes)



Role of atmospheric aerosols in the Global Climate

1. The Intergovernmental Panel on Climate Change (IPCC) concluded that emissions of greenhouse gases and aerosols due to human activities continue to alter the atmosphere in ways that are expected to affect the climate. Because aerosol particles in the atmosphere scatter sunlight back into space, they reduce the amount of energy that the planet absorbs, keeping it cooler. But this simplistic view ignored the fact that aerosol particles in the real world may also lead to heating of the lower atmosphere if they contain light absorbers such as elemental carbon and mineral dust. Scattering and absorbing the solar radiation is known as the “direct” effect of aerosols on the global climate, which can lead to either cooling or warming of the atmosphere.

2. Aerosols also have an “indirect” effect on climate by altering the properties of clouds, resulting in a change of their scattering properties. Indeed, if there were no aerosols in the atmosphere, clouds would be much less common. It is very difficult to form cloud droplets without small aerosol particles acting as “seeds” to start the formation of cloud droplets. Clouds with smaller droplets reflect more sunlight, and such clouds last longer, because it takes more time for small droplets to coalesce into droplets that are large enough to fall to the ground.

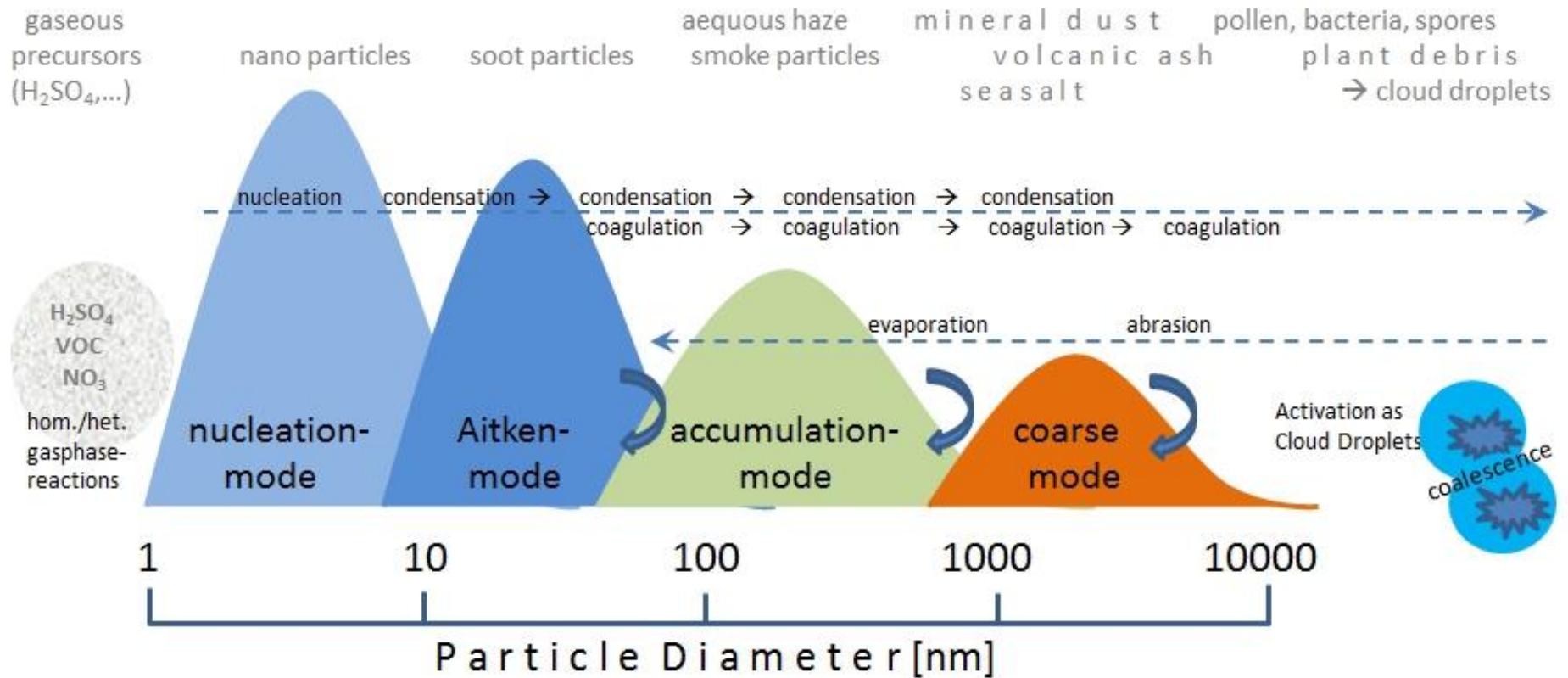
Particle Size Definition

Atmospheric particles are usually referred to as having a diameter, implying they are spherical. However, particles in the atmosphere have widely variable shapes for which geometrical diameters are often not meaningful. Some means of expressing the size of such particles is essential since many important properties of the particles such as volume, mass, and velocity depend on the size. In practice, the size of such irregularly shaped particles is expressed in terms of an equivalent diameter that depends on a physical, rather than a geometrical, property. An equivalent diameter is defined as the diameter of the sphere that would have the same value of a particular physical property as that of the irregular particle.

Particle Size Classification

Particle size is one of the most important parameters for characterising the behaviour of aerosols. The diameters of atmospheric aerosols span over four orders of magnitude, from a few nanometres to around 100 μm . To appreciate this wide size range one just needs to consider that the mass of a 10 μm diameter particle is equivalent to the mass of one billion 10 nm particles of the same composition. Atmospheric aerosols are normally classified into separate modes according to their size, formation process and atmospheric age.

Based on the number, surface, and volume distributions of aerosol particles four distinct groups of atmospheric particles can be defined.



1. Particles with diameters larger than $1.0\text{ }\mu\text{m}$ are identified as the coarse mode.

- These particles are mainly produced by mechanical processes and are introduced directly into the atmosphere from both natural and anthropogenic sources.
- The most significant source is the bursting of bubbles in the ocean, which creates coarse particles of sea salt.
- The wind also picks up dust, soil and biological particles and suspends them in the atmosphere.
- Anthropogenic coarse particles are introduced into the atmosphere through the abrasion of machinery, on the surface of roads, and in industrial and agricultural processes.

2. Particles with diameters between 0.1-1.0 μm are considered as the accumulation mode.

- A region of particle growth mainly due to the coagulation of particles with diameters smaller than 0.1 μm and from condensation of vapours onto existing particles, causing them to grow into this size range.
- They can also be introduced directly into the atmosphere, mainly through the incomplete combustion of wood, oil, coal, gasoline and other fuels.
- Because of the nature of their sources, particles in the accumulation size range generally contain substantial amounts of organic material as well as soluble inorganics such as ammonium, nitrate and sulphate.
- The accumulation mode is so named because particle removal mechanisms are least efficient in this regime, causing particles to accumulate there until they are ultimately lost through rain.

3. The Aitken mode particles, extending from 0.01 to 0.1 μm diameter.

- Formed from ambient-temperature gas-to-particle conversion as well as condensation of hot vapours during combustion processes.
- These particles act as nuclei for the condensation of low-vapour pressure gaseous species, causing them to grow into the accumulation range.
- The lifetime of these particles is short, as they are lost principally by coagulation with larger particles.
- The Aitken and accumulation mode particles are referred to collectively as fine particles.

4. The final type of particles are those with diameters smaller than $0.01\ \mu\text{m}$ and are known as ultrafine particles.

- They are thought to be generated by gas-to-particle conversion processes that are not yet well understood on a molecular level.
- Because of their very small size and mass, they are difficult to study given the available measurement techniques.
- These particles have been observed in bursts of very large numbers in the presence of either biogenic or anthropogenic emissions under favourable local conditions in many different environments including forests and coastal zones and have been referred to as the **nucleation mode**.

Particle size characterization

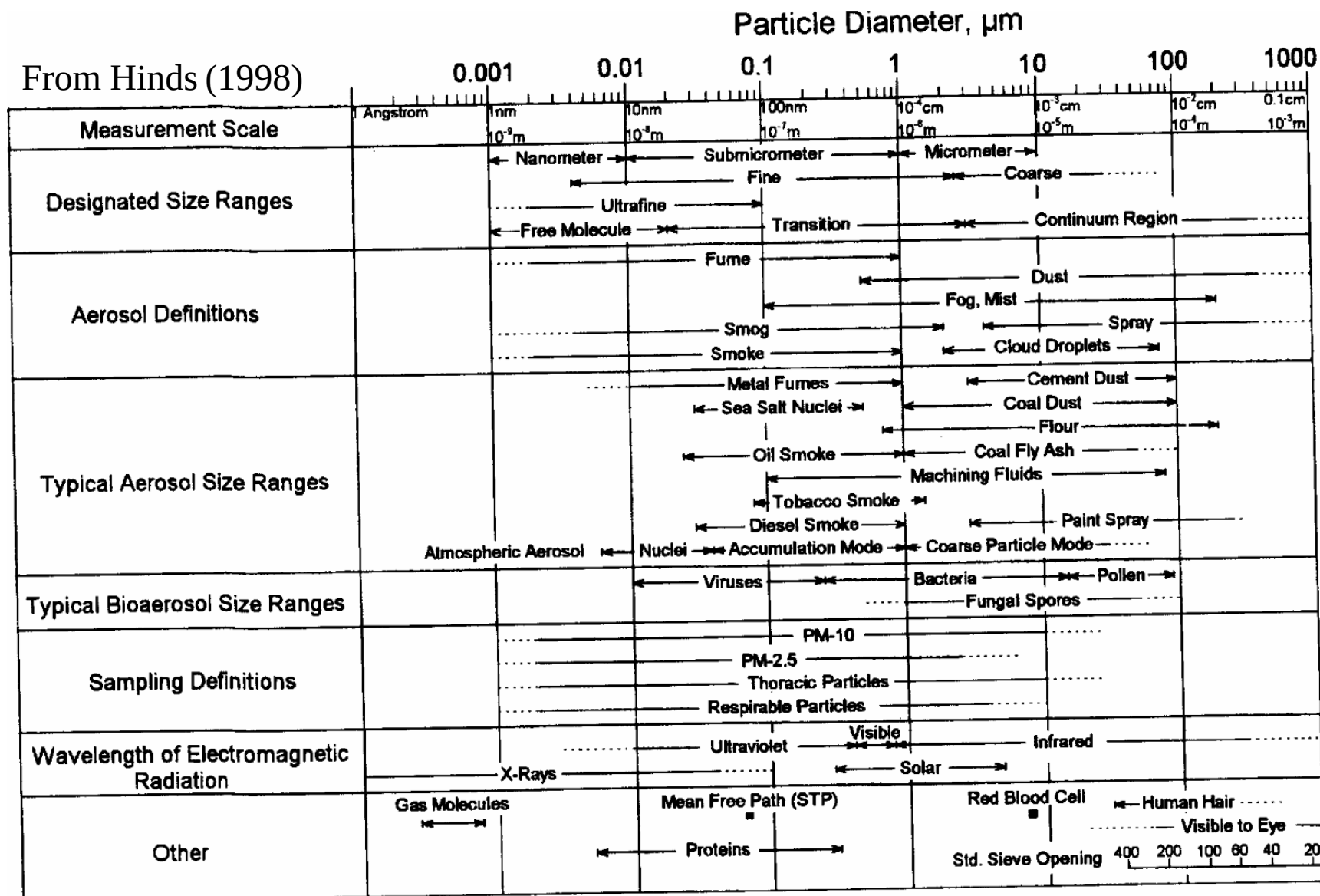


FIGURE 1.6 Particle size ranges and definitions for aerosols.